

ARRAYS

Q6) Maximize Items Bought Within Budget

DS	Session	Arrays	Question Information	Level 1 Challenge 26
Problem Description: Good news! Suresh get to go to America on a class trip! Bad news, he don't know how to use the Dollar which is the name of the American cash system. America uses coins for cash a lot more than the Kuwait does. Dollar comes in coins for values of: 1, 2, 10, 50, 100, & 500 To practice your Dollar skills, suresh have selected random items from Amazon.co.us and put them into a list along with their prices in Dollar. Suresh now want to create a program to check suresh Dollar math. Suresh goal is to maximize your buying power to buy AS MANY items as you can with your available Dollar. Input Format: File listing 2 to 6 items in the format of: ITEM DDDDD ITEM = the name of the item you want to buy DDDDD = the price of the item (in Dollar) Output Format: Print the output in a separate lines contains, List the items suresh can afford to buy. Each item on its own line. Suresh goal is to buy as many items as possible. If suresh can only afford the one expensive item, or 2 less expensive items on a list, but not all three, then list the less expensive items as affordable. If suresh cannot afford anything in the list, output "I need more Dollar!" after the items. The final line you output should be the remaining Dollar he will have left over after make purchases.				

CODE:

```
C main.c
1  #include <stdio.h>
2  #include <string.h>
3
4  struct Item {
5      char name[100];
6      int price;
7  };
8
9  int main(void) {
10     struct Item items[10];
11     int n = 0;
12     char line[200];
13
14     while (n < 6 && fgets(line, sizeof(line), stdin)) {
15         if (sscanf(line, "%s %d", items[n].name, &items[n].price) == 2) {
16             n++;
17         } else {
18             break;
19         }
20     }
21
22     int total_money = 1 + 2 + 10 + 50 + 100 + 500;
23
24     for (int i = 0; i < n - 1; i++) {
25         for (int j = 0; j < n - i - 1; j++) {
26             if (items[j].price > items[j + 1].price) {
27                 struct Item temp = items[j];
28                 items[j] = items[j + 1];
29                 items[j + 1] = temp;
30             }
31         }
32     }
33
34     int rem = total_money;
35     int bought = 0;
36
37     for (int i = 0; i < n; i++) {
38         if (items[i].price <= rem) {
39             printf("%s\n", items[i].name);
40             rem -= items[i].price;
41             bought++;
42         }
43     }
44
45     if (bought == 0) {
46         printf("I need more Dollar!\n");
47     }
48
49     printf("%d\n", rem);
50
51     return 0;
52 }
```

OUTPUT:

```
PS C:\Users\CH.SC.U4CSE25255\Projects\test> ./main
Book 50
Laptop 800
Pen 13
Headphones 200
^Z
Pen
Book
Headphones
400
```